

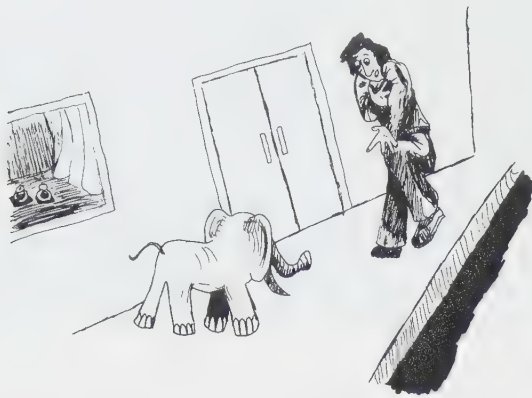


FIGURE 8.19 A “what-is-it?” picture, similar to those in one part of the Torrance Tests of Creative Thinking.

So, it doesn’t make sense to talk about “creative people”; it is better to talk about people who are “creative in a particular situation.” As a rule, people are creative only within the domains that they know well (Gardner, 1993). The key to creative success is primarily to generate many new ideas or products, more or less at random, and then judge which ones to develop and which to abandon (Simonton, 1997). Making that judgment requires extensive knowledge of the field. Therefore, you want a creative solution to a problem in automobile repair, try a good auto mechanic, not a painter or poet.

Howard Gardner (1993) studied creativity by examining in detail the lives of seven 20th-century people who are widely regarded as creative in very different fields: Sigmund Freud (psychology), Albert Einstein (physics), Pablo Picasso (painting), Igor Stravinsky (musical composition), T. S. Eliot (poetry), Martha Graham (dance), and Mahatma Gandhi (political resistance, religion, and spirituality). Gardner inquired what, if anything, these highly creative people had in common. He found a few patterns, which

may perhaps be accidents based on his particular choice of “creative people,” but these patterns deserve at least serious consideration. Among them were the following:

- Creativity thrives in an atmosphere of moderate tension; the person senses that the old ways of doing things are not quite right.
- The highly creative person must have enough of a background in the topic to feel self-confident but not so much experience that he or she has become trapped into the traditional habits of procedure. Just as people need about ten years to become an expert in a field, people generally need about ten years in a field before they are ready to make their first major creative contribution.
- During a period early in life, when a creative genius is working on some new ideas, he or she is likely to rely heavily on one or a small number of close, trusted friends for advice and encouragement. This advice is necessary for polishing the ideas, and the encouragement is necessary to enable the creative person to persist in the face of criticism.
- A creative genius throws him- or herself wholeheartedly into the work, sacrificing any possibility of a “well-rounded” life. Each of the creative people that Gardner studied had a very limited family life and very strained relationships with other people. Even Gandhi, who was famous as an advocate for love and justice, loved people in the abstract but had trouble developing close relationships with real, live people.

Creative careers, however, are extremely variable (Simonton, 1997). As a rule, poets are recognized for their greatness early in life, generally in their 20s, whereas the greatest, most creative historians seldom do their best work before their 40s or 50s. Within any field, some people start early and quit early and others bloom late; some—such as Bach, Picasso, and Thomas Edison—produce enormous quantities of good work, whereas others produce only one or two works worthy of recognition.



Howard Gardner studied the lives of seven highly creative people, including political and spiritual leader Mahatma Gandhi and dance pioneer Martha Graham, to find the features that promote creativity.

CONCEPT CHECK

5. It has been long reported that great creative poets die younger, on the average, than equally recognized historians. Why? (Hint: The average depends on the whole distribution, from youngest at death to oldest. Imagine the distribution of ages at death for both groups.) (Check your answer on page 295.)

Common Errors in Human Cognition

Although we humans pride ourselves on our intelligence and our ability to solve problems, we sometimes err with fairly simple problems. After someone points out the correct answer, we may be surprised by our own mistake. Sometimes we err because we relied on inappropriate heuristics. (Recall that heuristics are methods for simplifying a problem and facilitating an investigation.) Ordinarily, relying on heuristics enables us to find a reasonable, if not perfect, answer. Occasionally, however, certain heuristics can lead us astray. Let's now consider several reasons why people sometimes arrive at illogical conclusions.

Overconfidence

TRY IT YOURSELF

Let's start with a demonstration. Answer each of the following questions and then estimate the probability that your answer is correct. For example, you might feel 99%, 50%, or 10% sure that you are correct.

	Your answer	Percent confidence
1. What grade will you get in Psychology this semester?		
2. Who will be elected president of the United States in the next election?		
3. Over the next 3 months, will the stock market go up, down, or stay steady?		
4. What will happen in the next 2 years in Cuba?		
5. Will there be at least one hurricane and at least one earthquake somewhere in the world next year?		

Most people are overconfident of their answers to questions 1–4 and underconfident of their “yes” answer to question 5. (You did say “yes,” didn’t you?) For statistical reasons, the average person is almost sure to be underconfident of his or her answer to question 5: Because the actual probability is virtually 100%, no one will overstate the probability, and if even a few people guess less than 100%,

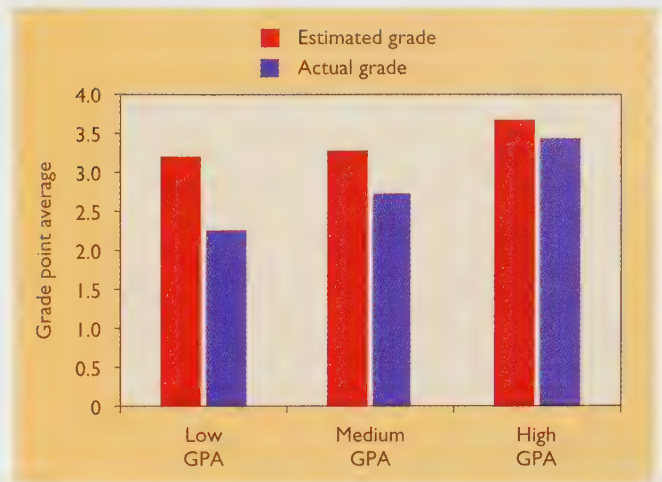


FIGURE 8.20 At the beginning of a semester, undergraduates in an advanced psychology course estimated their probable semester grade. Students with low, medium, or high grade-point averages generally predicted that they would get an A or a B. The best students were slightly overpredicting their success; the worst students were greatly overpredicting. (Based on data of Prohaska, 1994).

the average will be an understatement (Erev, Wallsten, & Budescu, 1994).

When people try to predict events that are actually quite uncertain, however, most people tend to be overconfident of their predictions (Juslin & Olsson, 1997). People are especially likely to be overconfident about their own accomplishments. For example, at the beginning of each semester, most students (even poor students) estimate that they will get mostly A's and B's (Prohaska, 1994). (See Figure 8.20.) Most students also slightly overestimate the grades they *already* received; that is, they remember slightly higher grades than they actually got (Bahrick, Hall, & Berger, 1996). Similarly, most athletes and coaches predict that they will have better seasons this year than they had last year, and most people entering a contest think they have a better-than-average chance of winning. A certain amount of optimism and self-confidence is probably good for our mental health, but objectively speaking, people overestimate their probable success. (For example, it is impossible for *every* team in the league to win more games this year than last year.)

Philip Tetlock (1994) conducted a study of government officials and consultants, foreign policy professors, newspaper columnists, and others who make their living by analyzing and predicting world events. He asked them to predict some events in U.S. and world politics over the next several years—such as what will happen in Korea, the Middle East, Eastern Europe, and Cuba—and to state their confidence in these predictions (such as 70%); 5 years later, he checked their accuracy. He found that most of the experts' predictions were no more accurate than the guesses you or I might have made, although they had stated great confidence in their predictions. Curiously, the few experts whose

predictions showed more than random accuracy shared a few common characteristics:

- They showed no strong political leaning. (Experts who consistently favored one political party or viewpoint were wrong as often as they were right.)
- They relied on complex information, instead of arguing from analogy to a single historical pattern.
- They were not overly sure of themselves. That is, the ones most likely to be right were the ones who admitted they might be wrong.

Of all the people who make their living by predicting the future, weather forecasters are among the most accurate. Although we tend to remember only their errors, of all the times when meteorologists foretell a “70% chance of rain,” it actually rains close to 70% of the time. One advantage of weather forecasting over, say, predicting political events, is that meteorologists get day-by-day feedback on their predictions; when they are wrong, they cannot avoid admitting they were wrong.

Premature Commitment to a Hypothesis

Sometimes we make mistakes because we commit ourselves prematurely to a particular hypothesis and fail to consider other possibilities. For example, examine the extremely out-of-focus photo in Figure 8.21a and guess what it depicts. Then see Figures 8.21b and 8.21c. Most people find that seeing the extremely out-of-focus photo makes it harder to identify the items in the slightly out-of-focus photo. When they saw the first photo, they formed a hypothesis, almost certainly a wrong one, which interfered with correct perception of the later photo (Bruner & Potter, 1964). This interference continues even after people are told that their first hypothesis was wrong (Snodgrass & Hirshman, 1991).

Peter Wason (1960) asked students to discover a certain rule he had in mind for generating sequences of numbers. One example of the numbers the rule might generate, he explained, was “2, 4, 6.” He told the students that they could ask about other sequences, and he would tell them whether or not those sequences fit the rule. As soon as they thought they had enough evidence, they could guess what the rule was.

Most students started by asking, “8, 10, 12?” When told “yes,” they proceeded with, “14, 16, 18?” Each time, they were told, “Yes, that sequence fits the rule.” Soon most of them guessed, “The rule is three consecutive even numbers.”

“No,” came the reply. “That is not the rule.” Many students persisted, trying “20, 22, 24?” “26, 28, 30?” “250, 252, 254?” And so forth. Eventually, they would say, “Three even numbers in which the second is two more than the first and the third is two more than the second.” Again, they were told that the guess was wrong. “But how can it be wrong?” they complained. “It always works!”

The rule Wason had in mind was, “Any three positive numbers of increasing magnitude.” For instance, 1, 2, 3,

would be acceptable; so would 5, 21, 24601. Where many students went wrong was in testing only the cases that their hypothesis said would fit the rule. One must also examine the cases that the hypothesis says will not fit the rule (Klayman & Ha, 1987).

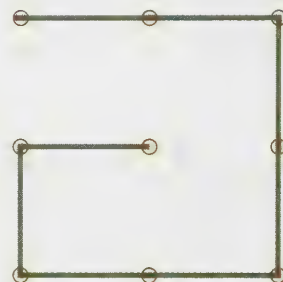
One special case of premature commitment to a hypothesis is the phenomenon of **functional fixedness**, the tendency to adhere to a single approach to a problem or a single way of using an item. Here are three examples:

TRY IT YOURSELF

1. You are provided with a candle, a box of matches, some thumbtacks, and a tiny piece of string that is shorter than the width of the candle, as shown in Figure 8.22. Using no other equipment, find a way to mount the candle to the wall so that it could be lit.
2. Consider an array of nine dots:



Connect all nine dots with a series of connected straight lines, such that the end of one line is the start of the next.



For example, one way would be:

But use the fewest lines possible.

3. There are some students in a room. All but two of them are psychology majors, all but two are chemistry majors, and all but two are history majors. How many students

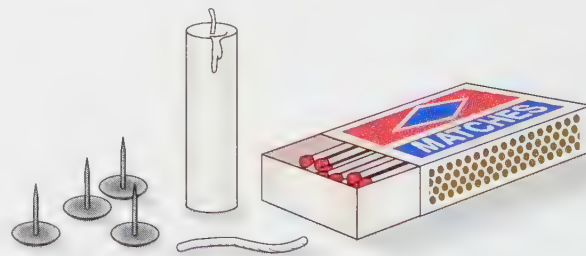


FIGURE 8.22 You are provided with a candle, a box of matches, some thumbtacks, and a very tiny piece of string. What is the best way, using no other equipment, to attach the candle to a wall?

are in the room, and what are their majors? (Note: If your first impulse is to say “two of each,” try it out: It doesn’t work.) Now here’s the interesting part: There are two possible solutions. After you have found one solution, discard it and find another.

After you have either found solutions to these questions or given up, check answer F on page 296. Question 1 was difficult because most people think of the matchbox as simply a container for the matches, not as a potential tool on its own. The box is “functionally fixed” for one way of using it. Question 2 was difficult because most people assume that the lines must remain within the area defined by the nine dots. Question 3 has a double difficulty: It is difficult to think of even one solution, and after thinking of it, it is hard to abandon it to think of an entirely different approach.

The Representativeness Heuristic and Base-Rate Information

Perhaps you have heard the saying: “If something looks like a duck, waddles like a duck, and quacks like a duck, chances are it’s a duck.” This saying is an example of the **representativeness heuristic**, *the tendency to assume that, if an item is similar to members of a particular category, it is probably a member of that category itself*.

This assumption is usually correct. It can lead us astray, however, if we are dealing with something rare. For example, suppose you see a bird that looks like an Eskimo curlew, walks like an Eskimo curlew, and whistles like an Eskimo curlew. Does that mean you have found an Eskimo curlew? Not likely. Eskimo curlews are extremely rare, almost extinct; chances are, you have sighted another bird that resembles an Eskimo curlew.

When we must decide whether something belongs in category A or category B—for example, an Eskimo curlew versus another bird—we should consider three questions: (1) How closely does it resemble the items in the first category? (2) How closely does it resemble the items in the second category? (3) Which is more common, category A or category B? The answer to the third question is known as **base-rate information**—that is, *data about the frequency or probability of a given item*, how rare or how common it is.

People frequently overlook the base-rate information and follow only the representativeness heuristic. As a result, they identify something as a member of an uncommon category, disregarding the more likely category. For example, consider the following question (modified from Kahneman & Tversky, 1973):

TRY IT YOURSELF

Psychologists have interviewed 30 engineers and 70 lawyers. One of them is Jack, a 45-year-old married man with four children. He is generally conservative, cautious, and ambitious. He shows no interest in political and social issues and spends most of his free time on home carpentry, sailing, and



FIGURE 8.21a People who form a hypothesis based on the first photo look at succeeding photos to find evidence that they are right. Because their first guess is generally wrong, they don’t do as well as people who look at the later photos before making any preliminary guesses. Try to guess what this shows. Then examine parts (b) and (c) on pages 291 and 293.

solving mathematical puzzles. What is the probability that Jack is one of the 30 engineers in the sample of 100?

Most people think that this description is more representative of engineers than it is of lawyers. Based on representativeness, they estimate that Jack is probably an engineer, neglecting to take into account that the sample includes more than twice as many lawyers as engineers. That base-rate information should influence their estimates. In fact, however, most people virtually ignore the base-rate information, making the same estimates regardless of whether the sample includes 30% or 70% engineers (Kahneman & Tversky, 1973).

Here is another example of overreliance on the representativeness heuristic. Read the following description, and then answer the questions following it:

Linda was a philosophy major. She is 31, bright, outspoken, and concerned about issues of discrimination and social justice.

Now, what would you estimate is the probability that Linda is a bank teller? What is the probability that she is a *feminist* bank teller? (Answer before you read on.)

The true probabilities, which are hard to estimate, are not the point. The interesting result is that most people estimate a higher probability that Linda is a *feminist* bank teller than the probability that she is a bank teller (Tversky & Kahneman, 1983). However, she could clearly not be a feminist bank teller without being a bank teller. Apparently, people regard this description as fairly typical for a feminist and therefore also for a feminist bank teller (or feminist

anything else) but do not regard the description as typical for bank tellers in general (Shafir, Smith, & Osherson, 1990). Because of the representativeness heuristic, people overestimate the probability that Linda is a feminist bank teller.

CONCEPT CHECK

6. Suppose an improved lie-detector test can determine with 90% accuracy whether people are telling the truth. An employer proposes to administer the test to all employees, asking them whether they have ever stolen anything valuable from the company and summarily firing everyone who fails the test. Is this policy reasonable? Assume that the company has 1,000 employees; of these, only 20 have ever stolen anything. Hint: Think about the base-rate probability of finding a dishonest employee. (Check your answer on page 295.)

The Availability Heuristic

When we are asked how common something is, we generally start by trying to think of examples. Try this question: In the English language, are there more words that start with *k* or more words that have *k* as the third letter? If you are like most people, you guessed that there are more words that start with *k*. How did you decide that? You tried to think of words that start with *k*: “king, kitchen, kangaroo, key, knowledge, . . .” Then you tried to think of words that have *k* as the third letter: “ask, ink, . . . uh . . .” You were relying on the **availability heuristic**, the strategy of assuming that the number of memories of an event that are available indicates how common the event actually is (Table 8.1). Because it was easier to think of words that start with *k* than words with *k* as the third letter, you assumed that there really are more words that start with *k*. In fact, however, words with *k* as the third letter are considerably more numerous.

The availability heuristic leads to illusory correlations, as we saw in Chapter 2. Someone asks, “Do people act strange on nights with a full moon?” If you have always expected people to act strange on such nights, you may be able to remember more times when they did act strange than times when they did not.

Here is another example of the availability heuristic: Suppose I ask you to fill out a survey of your beliefs on a variety of controversial political, social, and religious issues. Then, for each of those questions, I ask you to estimate what percentage of other people agree with your position. You will probably overestimate that percentage, for two reasons: One reason is that, as mentioned before, most people are overconfident of their own opinions. The other reason is that you probably associate with friends who share most of your opinions. Because you can think of many people who share your opinions, you overestimate how many others do also.

Another example: Do you consider yourself a better-than-average driver, an average driver, or a worse-than-average driver? Almost all people call themselves better than average or average in this regard. Where are all the below-average drivers? Evidently, some people overestimate their driving skills. One reason is this: When you think about other drivers, the ones you remember most vividly are the unusually bad drivers. Because the worst drivers are the easiest to remember, we overestimate their numbers.

You can guard against overuse of the availability heuristic. When you try to estimate whether one type of event is more common than another, look for systematic data. Don’t just trust your memory of how often various events occur.

Framing Questions

If we were truly logical beings, we would give the same answer to a question no matter how it was worded. In fact, most people give one answer to a question that is phrased in terms of gain but give a different answer to the same question when it is phrased in terms of loss.

For example, you have recently been appointed head of the Public Health Service. A new contagious disease has been detected, and you must choose between two plans of action. If you do nothing, 600 people will die. If you adopt plan A, you will save the lives of 200 people. If you adopt plan B, there is a 33% chance that you will save all 600 and 67% chance that you will save no one. (Choose plan A or B before reading further.)

Now another contagious disease breaks out; again you must choose between two plans. If you adopt plan C, 400

TRY IT YOURSELF

TABLE 8.1 The Representativeness Heuristic and the Availability Heuristic

	A TENDENCY TO ASSUME THAT . . .	LEADS US ASTRAY WHEN . . .	EXAMPLE OF ERRONEOUS ASSUMPTION
Representativeness heuristic	any item that resembles members of a particular category is probably itself a member of that category.	an item resembles members of a rare category.	You see something that looks the way you imagine a UFO would look, so you decide it is a UFO.
Availability heuristic	how easily we can think of examples of a category indicates how many examples really exist.	one kind of example is easier to think of than another is.	You remember more newspaper reports of airline crashes than of car crashes, so you assume that air crashes are more common than car crashes.

people will die. If you adopt plan D, there is a 33% chance that no one will die and a 67% chance that 600 will die. Choose one plan now, and then compare your choices with the results in Figure 8.23. *The tendency to answer a question differently when it is framed (phrased) differently is called the framing effect.*

Consider another example; this one deals with money instead of lives. Which would you rather have?

W. a gain of \$240

or X. a 25% chance to win \$1,000

Now you need to make another decision. You have just received an outright gift of \$1,000, but you must choose between two unpleasant alternatives:

Y. a loss of \$750

or Z. a 75% chance of losing the whole \$1,000
(a 25% chance of losing nothing)

Tversky and Kahneman (1981) found that 84% of all people chose W over X (avoiding risk), whereas 87% chose Z over Y (taking a risk). Note that W is actually \$10 less than choice Y and that X is the same as Z. Again, people generally avoid taking a risk when considering gains but accept a risk when considering losses. Put another way, people try to avoid losses.

The framing effect has consequences for persuasion. Suppose you want to encourage people to wear seat belts, exercise, take vitamins, quit smoking cigarettes, practice “safe sex,” or whatever. Should you say, “If you do this you



FIGURE 8.21b

will probably live longer”? Or should you say, “If you don’t do this, you will probably die sooner”? We need research to establish the results in each case, but the point is that we cannot assume that the “live longer” and “die sooner” messages will yield the same results (Rothman & Salovey, 1997). How we frame a question changes the way people respond to it.

Plan	How question was framed	Plan preferred by	Outcome
A	Save 200 people	72%	200 live, 400 die
C	400 people will die	22%	
B	33% chance of saving all 600; 67% chance of saving no one	28%	33% chance 600 live or 67% chance 600 die
D	33% chance no one will die; 67% chance all 600 will die	78%	

FIGURE 8.23 When Amos Tversky and Daniel Kahneman (1981) offered these choices to more than 150 people, 72% chose A over B, and 78% chose D over C. However, plan A is exactly the same as plan C (200 live, 400 die), and plan B is exactly the same as plan D. Why then did so many people choose both A and D? The reason, according to Tversky and Kahneman, is that most people avoid taking a risk when a question is phrased in terms of gain, but they are willing to accept a risk when the question is phrased in terms of loss.

CONCEPT CHECK

7. a. Someone says, “More than 90% of all college students like to watch late-late night television, whereas only 20% of older adults do. Therefore, more watchers of late-late night television are college students.” What error in thinking has this person made?
- b. Someone tells me that, if I say “abracadabra” every morning, I will stay healthy. I say it daily and, sure enough, I stay healthy. I conclude that saying this magic word really does ensure health. What error of thinking have I made? (Check your answers on page 296.)

Learning, Memory, Cognition, and the Psychology of Gambling

Suppose your professor asks everyone in class to hand in \$10. The professor collects all the money, keeps half of it, and hands the other half to one of the students, chosen at random. Presuming you were not the winner of the money, would you encourage your professor to repeat this exercise every week from now on?

Casino gambling and state lotteries operate much like this example. Table 8.2 illustrates the odds for one state lottery game. Actually, casinos offer much better odds than a state lottery; a casino keeps less than 5% of the gambled money, whereas a state government keeps about 50%. In either case, the great majority of the bettors are going to lose money, although they continue gambling anyway.

Gambling has been a common human behavior throughout history and throughout the world. Here we shall focus on why people make apparently illogical bets, where they are likely to lose more than they win. This issue illustrates several of the principles of learning, memory, and thinking we have discussed in the last three chapters.

TABLE 8.2 Expected Winnings on a \$1 Decco Ticket (a California Lottery Game)

PROBABILITY	×	PAYOFF	=	EXPECTED VALUE
.9573084	×	\$0	=	\$0.000
.040404	×	\$5	=	\$0.202
.0022409	×	\$50	=	\$0.112
.0000467	×	\$5,000	=	\$0.233
Total	1.0			\$0.547

NOTE: Someone who purchases a \$1 ticket has more than a 95% chance of winning nothing, slightly more than a 4% chance of winning \$5, and so forth. Overall, the person should expect to receive about 55 cents back for the \$1. No one actually receives the 55 cents for a single ticket, but that should be the average payoff after someone has bought many Decco tickets.



FIGURE 8.24 Most chronic gamblers believe that some people can win consistently at games of chance. Even with slot machines, many gamblers believe their skill in pulling the handle can influence their results.

Overestimation of Control

Rationally, you should not spend even \$1 when your chance of winning \$5,000 in a lottery is 1 in 21 thousand. But what if you thought that *your* chances were significantly better than everyone else's? Most gamblers believe that some people can win consistently at games of chance. Even most people who play the slot machines (Figure 8.24) believe that their skill in pulling the arm can influence their winnings (Griffiths, 1990).

In one study, people were given a chance to buy \$1 lottery tickets for a \$50 prize (Langer, 1975). Some of them were simply handed a ticket; others were permitted to choose their own ticket. Those who chose their own ticket thought they had a better chance of winning. Days later, all the ticket-holders were asked whether they were willing to sell their ticket to someone else. Those who had chosen their own tickets were much less willing to sell, and some refused to sell for less than the full \$50 they expected to win!

As a general rule, whenever we have an intention to achieve an outcome, we are inclined to assume that our actions will achieve that outcome, or at least make it more likely (Matute, 1996; Thompson, Armstrong, & Thomas, 1998). We are especially likely to believe that our actions matter in any competitive situation. Most of the time, our choices do influence the outcome, and healthy, successful people keep trying and believing that they are succeeding. That tendency backfires, however, in gambling situations where the outcomes are actually random.

People's Affinity for Long-Shot Bets

TRY IT YOURSELF

Which would you rather have, \$100,000 for sure, or a 10% chance of winning a million dollars? What about a choice between \$10,000 for sure and a 1% chance at a million dollars? Would you prefer

\$1,000 for sure or a 0.1% chance at a million? A sure profit of \$100 or a 0.01% chance at a million? A \$10 profit or a 0.001% chance at a million? A \$1 profit or a 0.0001% chance at a million?

If you are like most people, you chose the \$100,000 over a 10% chance at a million, and you also chose \$10,000 over a 1% chance. But at some point, you switched from the sure profit to the gamble. Especially if the sure profit is \$10 or less, most people prefer a chance at winning a million. In fact, almost half of college students who were asked would forego \$10 to have even one chance in a million of winning a million—a gamble of very bad odds (Rachlin, Siegel, & Cross, 1994).

Why? First, for most students, \$10 does not sound like very much. Oh, sure, you would rather have \$10 than not have it, but another \$10 is not going to raise most people's standard of living enough to notice. A million dollars would. Furthermore, although it is easy to understand the difference between a 10% chance and a 1% chance, most of us are not very concerned about the difference between a 0.001% chance and a 0.0001% or even a 0.000001% chance. Beyond a certain point, they all blur together as just "unlikely," and as long as the chance is even slightly above 0, we think, "Someone is going to win, and it might be me." Finally, it may be that the extreme unlikeliness of winning makes the bet even more appealing: People report more pleasure from a surprising gain than from an expected gain (Mellers, Schwartz, Ho, & Ritov, 1997). That is, a long-shot gambling win, which is a big surprise, provides more fun than your paycheck at work, or the money you knew your aunt would send you on your birthday, or any other routine gain.

Schedules of Reinforcement

Anyone who makes many bets will win some and lose some. The effect is similar to a variable-ratio schedule, as discussed in Chapter 6, but not exactly the same. (In a variable-ratio schedule, an unpredictable number of responses will eventually lead to reinforcement. In gambling, it is possible to keep on betting forever without winning.) Intermittent schedules of reinforcement lead to continued responding, even during a long period without reinforcement.

Howard Rachlin (1990) suggests that gamblers evaluate their success at the end of a string of bets that ends in a win. For example, suppose you lose bets on six horse races and then bet on a winner. If your bet on the winner pays more than the amount you lost on the first six bets, then you don't count your progress as "six losses, one win," but as "a net gain on the day, one overall win."

Now, imagine a man who buys tickets in a state lottery. Every year he spends hundreds of dollars, maybe thousands, without winning anything. Does he give up? If so, he must admit that all he had invested was a loss. If he continues to buy tickets, he maintains the hope of eventually hitting the jackpot, winning more than enough to make up



FIGURE 8.21c

for all the money he lost. One single payoff could make the whole string of bets a net win.

Vicarious Reinforcement and Punishment

Recall from the discussion of social-learning theory that people learn what to do and what to avoid by observing what happens to others. Recall also that vicarious reinforcement tends to be more effective than vicarious punishment.

State governments want to encourage people to buy lottery tickets (because they provide revenue for the government), so they encourage massive publicity for everyone who wins a big jackpot (Figure 8.25). You often see news reports



FIGURE 8.25 States that sponsor lotteries provide publicity and an exciting atmosphere for each big payoff. They hope that this publicity will provide vicarious reinforcement to encourage other people to buy lottery tickets. They do not publicize all the people who lost money on the lottery.



Gambling occurs in almost every culture that has possessions and money. Here, Egyptian men place bets at a racetrack.

showing some instant millionaire, delirious with excitement. (The state hopes that this vicarious reinforcement will induce you to buy lottery tickets.) You seldom see reports about the millions of people who bought tickets and won nothing.

The Influence of Heuristics

Using the availability heuristic, people assume that, if they can recall many examples of an event, that event must be common. You can see how this combines forces with vicarious reinforcement. Someone who sees a big lottery winner on television almost every week remembers many such examples and therefore overestimates the likelihood of winning the lottery.

The representativeness heuristic also plays a role in gambling. Suppose you are flipping a coin, recording the order of heads (H) and tails (T). Which of the following sequences do you think is more likely?

1. H H H H H H H H
2. H H H H H H H T

Although both sequences are equally likely, most people who have no training in probability theory think sequence 2 is more likely (Kahneman & Tversky, 1973). A sequence of all heads seems *unrepresentative* of usual experiences. However, representativeness does not alter the odds: Even after seven heads, another head is just as likely as a tail on the next flip.

The **gambler's fallacy**, an example of the representativeness heuristic, is *the belief that, if a particular outcome has not occurred for a while, its "turn" has come*. With regard to human behavior, this assumption usually succeeds: If your friends have gone out for pizza four nights in a row, they are indeed due to switch to something else. But truly random events do not "take turns." If the last seven spins on the roulette wheel landed on black numbers, red does not become more likely for the next spin. Losing your last 20 bets does not make you more likely to win your next bet.

Self-Esteem

Many gamblers have an additional reason for gambling that we cannot explain in terms of learning, memory, or errors in reasoning: They want to beat their opponent. For many habitual gamblers, their self-esteem is strongly tied to proving themselves to be "winners," and they like to make that point by winning someone else's money (Peck, 1986). Their self-esteem becomes more of an issue when gambling among friends and acquaintances than it does when gambling in a casino. Consequently, a habitual gambler who has been losing at a game of, say, craps or poker will sometimes bet wildly and foolishly in an effort to catch up.

In summary, we have the following explanations for why some people continue to gamble despite consistent losses:

- They believe that by skillfully choosing the right numbers they can increase their probability of winning a game of chance.
- If the potential prize is large, people act as if they do not understand the difference between a small chance of winning and an *extremely* small chance of winning.
- They can continue through a long losing streak if the (imagined) eventual payoff would be large enough to repay all their losses.
- They remember seeing or hearing about many people who have won big jackpots. They thus receive vicarious reinforcement for gambling, and they overestimate the probability of winning.
- They may believe that, after a long series of losses, the probability of winning increases.
- They are so eager to beat an opponent that they will take risks they know to be foolish.

Similar explanations apply to the gambles that people take in everyday life. If you drink and drive, or date someone who mistreated you in the past, you are taking a gamble—perhaps a foolish gamble. People take such risks for some of the same reasons just described.

One general point is that people usually have multiple reasons for their behavior. People do not gamble for just one reason any more than you went to college for just one reason. A second general point is that the principles of learning, memory, and cognition do not apply to separate domains; we can apply all of those principles to a single behavior, such as gambling.

SOMETHING TO THINK ABOUT

Recall the discussion on page 290 about how the phrasing of a question can influence someone's answer. For example, most people will take more risks to avoid a loss than they will to increase a gain. Can you use this principle to explain why many gamblers on a losing streak will continue betting, sometimes increasing their bets? Is there a different way for a gambler to think about the situation, to decrease the temptation to continue gambling? ★

THE MESSAGE

Successful and Unsuccessful Problem Solving

After reading about all the common reasoning errors people make and the foolish gambles that people take, it is easy to despair of human intelligence and assume that we are all a bunch of fools. In a way we are, but in a way we are all geniuses too. Each of us has areas of expertise and near-expertise where we make excellent decisions, and areas of incompetence where we make embarrassing blunders. Learning about common errors may help us to avoid these blunders . . . at least sometimes.

SUMMARY

- ★ **Becoming an expert.** Experts are made, not born. Becoming an expert requires years of practice and effort. (page 279)
- ★ **Expert pattern recognition.** Experts recognize and memorize familiar and meaningful patterns more rapidly than less experienced people do. (page 281)
- ★ **Steps for solving a problem.** People take four steps to solve a problem: understanding the problem, generating hypotheses, testing the hypotheses, and checking the result. (page 281)
- ★ **Algorithms and heuristics.** People can solve problems by using algorithms (repetitive means of checking every possibility) or heuristics (ways of simplifying the problem to get a reasonable solution). (page 282)
- ★ **Generalizing.** People who have learned how to solve a problem will not necessarily apply that solution to a similar problem. (page 283)
- ★ **Insight.** With insight problems, people have trouble estimating how close they are to a solution. However, they may be making progress even if they do not realize that they are. (page 284)
- ★ **Creativity.** Creative people do their creative work in a field in which they have knowledge and self-confidence; creativity does not generalize well from one field of endeavor to another. Many highly creative people go through a period in which they rely on a small group of friends, perhaps just one, for support and encouragement. They dedicate their lives to their work, often to the exclusion of all else. (page 285)
- ★ **Reasons for errors.** People tend to be overconfident about their own judgments. Other common mistakes in human reasoning include premature commitment to a hypothesis, overreliance on the representativeness heuristic and the availability heuristic, and altering one's answer when the same question is framed in different terms. (page 287)
- ★ **Reasons for gambling.** Gambling illustrates the combined influences of learning, memory, and cognition. Many people gamble despite consistent losses, because they overestimate their control, because they are insensitive to the difference between a small chance and an extremely small chance

of winning, because they experience vicarious reinforcement, and because they believe a series of losses makes an eventual win more likely. Some people gamble as a way of competing with others and of establishing their own self-esteem. (page 292)

Suggestion for Further Reading

Bransford, J. B., & Stein, B. S. (1984). *The ideal problem solver*. New York: Freeman. Advice about how to approach and solve both “mind-bender” problems and practical problems.

Terms

- algorithm** a mechanical, repetitive mathematical procedure for solving a problem (page 282)
- heuristics** strategies for simplifying a problem or for guiding an investigation (page 282)
- creativity** the development of novel, socially valued products (page 285)
- functional fixedness** the tendency to adhere to a single approach to a problem or a single way of using an item (page 288)
- representativeness heuristic** the tendency to assume that, if an item is similar to members of a particular category, it is probably a member of that category itself (page 289)
- base-rate information** data about the frequency or probability of a given item (page 289)
- availability heuristic** the strategy of assuming that the number of available memories of an event indicates how common the event actually is (page 290)
- framing effect** the tendency to answer a question differently when it is framed (phrased) differently (page 291)
- gambler's fallacy** the belief that, if a particular outcome has not occurred for a while, it becomes more likely than before (page 294)

Answers to Concept Checks

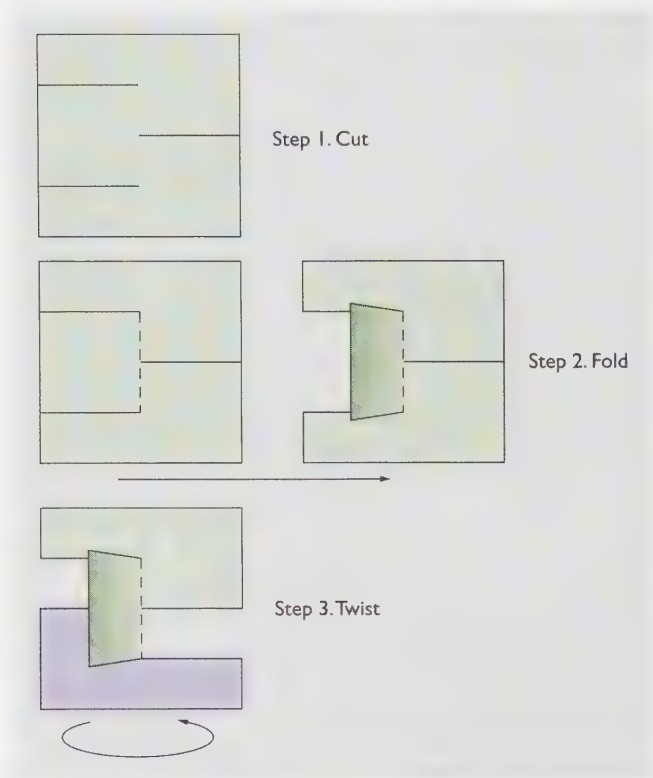
4. An algorithm would check each of the possible routes: If you had your home city (H) and three other cities (1, 2, and 3), the possible routes would be H-1-2-3-H, H-1-3-2-H, and H-2-1-3-H. (Three other routes that you could generate would be the mirror images of these three and therefore not necessary to consider.) As the number of cities increases, the number of possible routes rises rapidly. If you had to visit 10 cities, your algorithm would have to consider almost 2 million routes. One possible heuristic would be to consider only those routes from each city to one of the closest two other cities. (page 283)
5. Because many poets are recognized for their greatness while they are still young, it is possible to be a great poet and die young. As a rule, historians are in their 40s or later when they do their first outstanding work; therefore, it is almost impossible to be recognized as a great historian and to die young. (page 287)
6. The employer would fire 18 dishonest employees (90% of the 20 who had stolen). The employer would also fire 98 honest employees (10% of the 980 who had not stolen). That is, a clear majority of those identified as dishonest are

actually honest. (An employer might be willing to fire that many honest people in order to get rid of some dishonest people, but most employees would consider this action most unfair.) (page 290)

7. a. Failure to consider the base rate: 20% of all older adults is a larger number than 90% of all college students;
b. Premature commitment to one hypothesis without considering other hypotheses (such as that one could stay healthy without any magic words). (page 292)

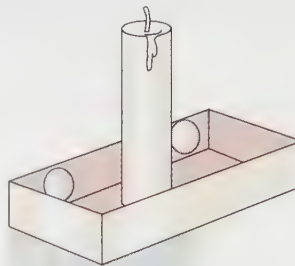
Answers to Other Questions in the Text

- C. The water in the tube would leak out of the hole in the bottom. Any membrane heavy enough to keep the water in would also keep the rubber balls out. (page 283)
D. This illustration shows how to cut and fold an ordinary piece of paper or cardboard to match the figure, with nothing left over. (page 284)

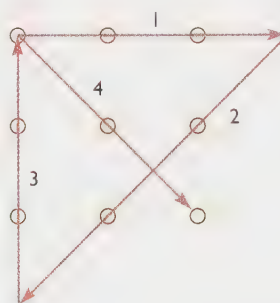


- E. Set 1: The words *playing*, *credit*, and *report* are all associated with *card*. Set 2: The words *ticket*, *shop*, and *broker* are all associated with *pawn*. (page 285)

- F. (1) The best way to attach the candle to the wall is to dump the matches from the box and thumbtack the bottom of the box to the wall, as shown in this picture. The tiny piece of string is useless.



- (2) The dots can be connected with four lines:



- (3) One possible answer is that there are three students: one psychology major, one chemistry major, and one history major. The other possibility is that there are two students who are majoring in something else—music, for example. (If there are two music majors, all but two of them are indeed majoring in psychology, etc.) (page 288)

Web Resources

Creativity and Star Trek

www2.shore.net/~sek/STCreativity.html

Robert Sekuler and Randolph Blake, authors of *Star Trek on the Brain: Alien Minds, Human Minds*, offer an interesting exploration of creativity and some of the limitations on human creativity. Before you visit the creativity page, take a few minutes to draw a picture of a member of an alien race that no human has ever seen. Make it as different from human as you can. Click the link to the "Star Trek on the Brain" page for several interesting links.

Language

How do we learn to understand language, to speak, and to read?

Does language have special properties that set it apart from other intelligent behaviors?

Language is an extremely versatile system. Other species have various ways of sending signals to one another, but only human languages have the property of **productivity**, *the ability to express new ideas*. Every day we say and hear a few stock sentences, such as “Nice weather we’re having,” or “I can’t find my shoes,” but we also say and hear sentences that no one has ever said before.

You might ask, “How can you know that no one has ever said that particular sentence before?!” Well, of course, we cannot know for sure that a particular sentence is new, but we can be confident that many sentences are new (without specifying which ones). Imagine trying this exercise (don’t really try it unless you have an enormous amount of time to spare): Pick a sentence of 10 to 20 words from any book you choose. How long would you need to keep reading, in that book or any other, until you found exactly the same sentence again?

In short, we do not memorize all the sentences we will ever use; instead, we learn rules for making sentences and for interpreting other people’s sentences. The famous linguist Noam Chomsky (1980) has described those rules as a **transformational grammar**, which is *a system for converting a deep structure into a surface structure*. The deep structure is the underlying logic of the language. The surface structure is the sequence of words as they are actually spoken or written (Figure 8.26). According to this theory, whenever we speak we transform the deep structure of the language into a surface structure. Two surface structures can resemble each other without representing the same deep structure, or conversely, they can represent the same deep structure without resembling each other.

For example, “John is easy to please” has the same deep structure as “pleasing John is easy” and “it is easy to please John.” These sentences all represent the same underlying idea.

In contrast, consider the sentence, “Never threaten someone with a chain saw.” The surface structure of that sentence maps into two quite different deep structures:

It is not nice to swing a chain saw around and threaten someone with it.

If you meet someone carrying a chain saw, don’t make any threatening gestures.

The productivity of language enables humans to communicate and elaborate on ideas to a far greater extent than any other species can. Language researcher Terrence Deacon describes a brief talk about language and the brain that he presented to his 8-year-old son’s elementary school class. One child in the class asked whether other animals have their own languages. Deacon explained that other species have various methods of communication, but none of them has a flexible system similar to human language. The child persisted, asking whether other animals had at least a *simple* language, perhaps one with only a limited vocabulary and a very simple grammar. No, he replied, they do not have even a simple language.

Then another child asked, “Why not?” (Deacon, 1997, p. 12). Deacon paused. And then paused some more. Why not, indeed. He realized that this 8-year-old child had asked a most profound question. If language is so extremely useful to humans, why haven’t other species evolved at least a little of it?

Language comes easily to humans. Almost all children, even many mentally retarded children, learn their native language, even though their parents may know nothing about how to teach it. Meanwhile, nonhumans can learn only a little language at best, despite psychologists’ concentrated efforts to teach it. Evidently, we humans are highly specialized for acquiring language.

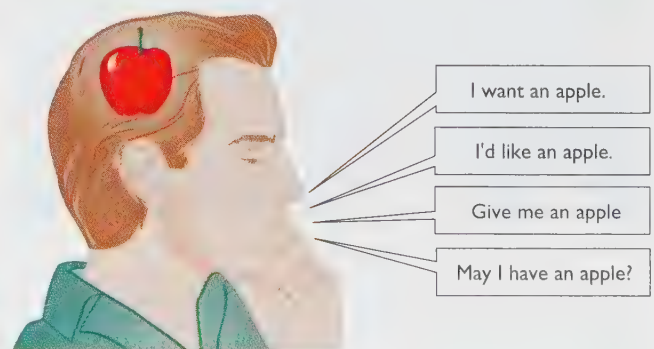


FIGURE 8.26 According to transformational grammar, we can transform a sentence with a given surface structure into any of several other sentences with different surface structures. All of them represent the same deep structure, which is the underlying logic of the sentence.

Precursors to Language in Chimpanzees and Bonobos

One way to examine humans' specialization for language is to determine how far other species could progress toward language, given sufficient training. Beginning in the 1920s, several psychologists reared chimpanzees in their homes and tried to teach them to talk. The chimpanzees learned many human habits (Figure 8.27) but showed only very limited understanding of language. Later, Allen Gardner and Beatrice

Gardner (1969) taught a chimpanzee named Washoe to use the sign language that American deaf people use (Ameslan). Sign language is closer to the hand gestures that chimpanzees use naturally, and it does not require them to imitate human voice sounds, for which their vocal tracts are poorly adapted. Washoe eventually learned the symbols for about 100 words, which she occasionally linked into meaningful combinations, such as “cry hurt food” for a radish.

Other chimpanzees also learned to use such symbols, but a controversy arose about whether their use of symbols and gestures really resembles human language. In many



a



b



c



d



e

FIGURE 8.27 Speaking is physically impossible for chimpanzees, but some psychologists have tried to teach them to communicate by using gestures or symbols. (a) One of the Premacks' chimps arranges plastic chips to make a “sentence” request for food. (b) Viki in her human home, helping with the housework. After years with the Hayeses, she could make only a few sounds similar to English words. (c) Kanzi, a bonobo, presses symbols to indicate words. Among the primates, bonobos have shown the most promising ability to acquire language. (d) A chimp signing *toothbrush*. (e) Roger Fouts with Alley the chimp, who is signing *lizard*.

cases, chimpanzees' gestures were an imitation of symbols their human trainers had recently used (Terrace, Petitto, Sanders, & Bever, 1979). Moreover, they used their symbols almost exclusively to make requests, only rarely to describe things, and they seldom linked gestures together in anything like a sentence (Pate & Rumbaugh, 1983; Terrace et al., 1979; Thompson & Church, 1980). By contrast, a human child with a vocabulary of 100 words or so starts to link them together to make many original combinations and short sentences.

That, at any rate, was the conclusion derived from studies of the common chimpanzee, *Pan troglodytes*. Closely related to the common chimp is a rare and endangered species, *Pan paniscus*, sometimes known as the pygmy chimpanzee (a misleading term because these animals are almost as large as common chimpanzees) and sometimes known as the bonobo. Their social behavior resembles that of humans in several regards: Males and females form strong, long-lasting attachments; females are sexually responsive throughout the month, not just during their fertile period; males contribute much more to infant care than do other nonhuman primate males; and adults often share food with one another.

Several bonobos have used symbols in impressive ways. First, they occasionally use the symbols to name and describe objects that they are not requesting. Second, they sometimes use the symbols to describe past events. (One explained a cut on his hand by using gestures to indicate that his mother had bit him.) Third, they frequently make original, creative requests, such as asking one person to chase another person while the bonobo watched. Their comprehension is better than their production; a couple of bonobos seem to comprehend symbols about as well as a 2- to 2½-year-old child understands language (Savage-Rumbaugh et al., 1993).

A couple of bonobos have also shown considerable understanding of spoken English (Figure 8.28), responding correctly to unfamiliar spoken commands such as “throw your ball into the river” and “go to the refrigerator and get out a tomato” (Savage-Rumbaugh, 1990; Savage-Rumbaugh, Sevcik, Brakke, & Rumbaugh, 1992). They even passed the test of responding to commands issued over earphones, used to eliminate the possibility of unintentional “Clever Hans”-type signals, as discussed in Chapter 2.

The explanation for this impressive success probably pertains partly to species differences: Perhaps bonobos have greater language capacities than common chimpanzees. Another part of the explanation may pertain to the method of training: Learning by observation and imitation probably promotes better understanding than the formal training methods that were used in previous studies. (Savage-Rumbaugh, Sevcik, Brakke, & Rumbaugh, 1992). Finally, the bonobos began their language experience early in life. Songbirds learn their song more easily during an early sensitive period; humans apparently have a sensitive period for easy language acquisition early in life; perhaps the same is true for bonobos.



FIGURE 8.28 Kanzi, a bonobo, points to answers on a board in response to questions he hears through earphones. Experimenter Rose Sevcik sits with him but does not hear the questions, so she cannot intentionally or accidentally signal the correct answer.

CONCEPT CHECK

8. Based on the studies with bonobos, can you offer some good advice about how to teach language to children born with a disability that impairs language learning? (Check your answer on page 311.)

Human Specializations for Learning Language

Is the glass half full or half empty? Should we be impressed that bonobos understand language almost as well as a 2½-year-old child, or wonder why they do not progress further? Yes, bonobos sometimes construct two- or three-symbol combinations, but they do not produce full sentences. Humans are clearly specialized to learn language in a way that no other species can.

Susan Carey (1978) has estimated that children between the ages of 1½ and 6 learn an average of nine new words per day—almost one new word per hour! Deciphering the meanings of words would seem to be an overwhelmingly difficult task (Markman, 1990). Suppose someone points to a pillow and says “makura.” If you were a Japanese child, you would accept *makura* as the word for *pillow*. You would also accept it as an adult trying to learn Japanese as a second language. But how did you know that the word *makura* meant *pillow*? Logically, it could have referred to all soft things, all cloth-covered things, all throwable things, or all things important at bedtime (including pajamas, darkened rooms, storybooks, and goodnight kisses). Sometimes children do guess wrong. For example, if you point to a dog and say “dog,” most young children assume that the word also

applies to other small animals. Similarly, many young children first use the word *daddy* to refer to all adult men. But it is remarkable that, when children learn new words, they more or less understand the meanings.

Noam Chomsky has argued that language is so complex, and so easily learned, that its development must not depend on ordinary types of learning. For example, consider this sentence: *The boy who is unhappy is watching Mickey Mouse*. How would you convert that sentence to a question, to ask whether it was true? I assume you find this easy: "Is the boy who is unhappy watching Mickey Mouse?" (You might also have asked, "Is the boy who is watching Mickey Mouse unhappy?") Even most preschool children phrase the question correctly; virtually none of them say, "Is the boy who unhappy is watching Mickey Mouse?" How did those preschoolers know the right way to phrase the question? Chomsky argues that children have so seldom heard such questions that they have not had adequate opportunity to learn the rather complicated grammatical rule. He and his followers therefore suggest that people are born with a **language acquisition device** or "language instinct," *a built-in mechanism for acquiring language* (Pinker, 1994). The basis for this theory is known as the **poverty of the stimulus argument**: *What children hear does not provide enough information for them to learn or infer grammar, so they must be born with that knowledge.*

Most psychologists concede that people must be born with some sort of language predisposition; after all, children who are taught under the worst conditions acquire far more language than chimpanzees taught under the best conditions. However, many researchers still question the poverty of the stimulus argument, because grammar differs enormously from one language to another. If you grew up in an English-speaking family, you quickly learned to phrase questions like, "Is the boy who is unhappy watching Mickey Mouse?" But if you had been adopted in infancy by a family living somewhere else in the world, you could have learned to speak Finnish, Mandarin, Hindi, Navajo, Zulu, or any other language, which might have a grammar radically different from English. It is easy to agree that infants must be born with something that facilitates language learning, but it is implausible to suppose that this something would "tell" the infant how to phrase an English question about an unhappy boy who might be watching Mickey Mouse (Deacon, 1997; Seidenberg, 1997).

We are forced, then, to reexamine the idea that infants can actually learn all the complexities of word meanings and grammar from apparently meager information. Perhaps this task is not as nearly impossible as it sounds. In one study, researchers fed the text of an electronic encyclopedia into a computer that was programmed to detect which words were used in similar contexts. At the end, the computer was able to pass simple language tests, such as identifying synonyms for a word (Landauer & Dumais, 1997). Furthermore, parents throughout the world simplify language by speaking "parentese." I am not talking about idiotic "goo-goo" baby

talk, but a pattern of speech that emphasizes and prolongs the vowels, making more clear than usual the difference between words such as *cat* and *cot* (Kuhl et al., 1997). Infants listen more intently to *parentese* than to normal speech and learn more from it.

Some psychologists have even proposed that language learning is easy for children specifically *because* their memory is imperfect (Newport, 1990). They cannot remember everything they hear, so they remember the patterns that occur regularly; in effect, they ignore the trees and perceive the forest.

Language and the Human Brain

What aspect of the human brain enables us to learn language so easily? One hypothesis is that language is simply an accidental by-product of humans' large brains and superior intelligence. One major problem with that hypothesis is that dolphins and whales have larger brains than humans have but do not develop language. (Yes, they communicate, but not in a flexible system resembling human language.) Furthermore, some people with massive brain damage have less total brain mass than a chimpanzee, but depending on the location of the damage, they can continue to speak and understand language. Also, people with certain genetic abnormalities develop normal intelligence except for impaired language (Gopnik & Crago, 1991), or impressive language skills despite mental retardation in all other regards (Bellugi, Wang, & Jernigan, 1994). So language is not just a by-product of overall intelligence or brain size; it must reflect a special organization of our brain.

Studies of brain-damaged people have long pointed to two brain areas as particularly important for language. People with damage in the frontal cortex, including *Broca's area* (Figure 8.29), develop **Broca's aphasia**, *a condition characterized by inarticulate speech and by difficulties with both using and understanding grammatical devices—prepositions, conjunctions, word endings, complex sentence structures, and so forth.* These people do not really lose all grammatical understanding; they merely find it much more difficult to use and understand language, much as undamaged people do when they are extremely distracted (Blackwell & Bates, 1995). People with damage in the temporal cortex, including *Wernicke's area* (Figure 8.29), develop **Wernicke's aphasia**, *a condition marked by difficulty recalling the names of objects and impaired comprehension of language.* Because these people do not remember names, their speech is nonsensical, even when it is grammatical. For example, one patient responded to a question about his health, "I felt worse because I can no longer keep in mind from the mind of the minds to keep me from mind and up to the ear which can be to find among ourselves" (Brown, 1977).

However, language did not evolve by simply adding a language circuit to an otherwise nonlinguistic brain. The

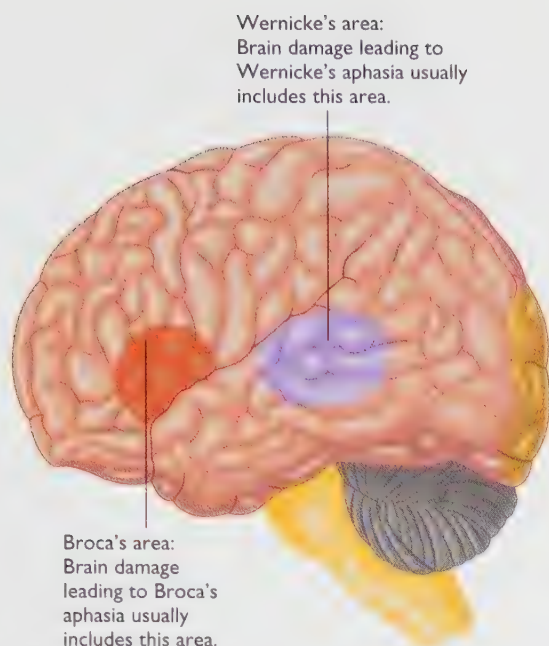


FIGURE 8.29 Brain damage that produces major deficits in language usually includes the left-hemisphere areas shown here. However, the deficits are severe only if the damage is more extensive, including these areas but extending to others as well. Many areas of the human brain contribute to language comprehension and production.

human brain is not just an ape brain with language ability tacked on; it displays enormous expansion of the prefrontal cortex, increased connections among cortical areas, and numerous other specializations to facilitate language (Deacon, 1997). Broca's area and Wernicke's area are important for language, but they are hardly the only important areas. Brain damage that seriously impairs language generally includes either Broca's or Wernicke's area but also includes other areas. Furthermore, PET scans or other recordings of the brain show widespread activation during speech or

language listening. It is hardly an exaggeration to say that the whole human brain is specialized to use and manipulate symbols to make language possible.

Stages of Language Development

Table 8.3 lists the average ages at which children reach various stages of language ability (Lenneberg, 1969; Moskowitz, 1978). Progression through these stages depends largely on maturation, not just extra experience (Lenneberg, 1967, 1969). Parents who expose their children to as much language as possible find that they can increase the children's vocabulary a bit, but they hardly affect the ages at which children reach various stages of language development (Figure 8.30). At the other extreme, in terms of exposure to language, hearing children of deaf parents are exposed to very little spoken language, but if they have periodic contact with speaking people, they progress through the various stages of language development almost on schedule.

For the first 6 months or so, deaf infants babble as much as hearing infants do. After that, deaf infants babble less. At first, babbling produces only haphazard sounds, but soon the infant starts repeating the sounds that are common in the language that the baby has been hearing, while discarding other sounds that the baby has not been hearing. Thus, by the age of 1 year, French babies babble French sounds, Chinese babies babble Chinese sounds, and so forth (Locke, 1994).

By the age of 1 year, infants begin to understand language, and most can say at least a word or two. Throughout the world, one of the first sounds to be uttered is *muh*. Parents in most parts of the world have defined *muh-muh* (or something similar to it) as meaning "mother." Infants also typically make the sounds *duh*, *puh*, and *buh*; they almost never make the sound *s*. In many languages, the word for father is similar to *daddy* or *papa*. *Baba* is the word for grandmother in several languages. In effect, infants tell

TABLE 8.3 Stages of Language Development

AGE	TYPICAL LANGUAGE ABILITIES (MUCH INDIVIDUAL VARIATION)
3 months	Random vocalizations.
6 months	More distinct babbling.
1 year	Babbling that resembles the typical sounds of the family's language; probably one or more words including "mama"; language comprehension much better than production.
1½ years	Can say some words (mean about 50), mostly nouns; no phrases.
2 years	Speaks in two-word phrases.
2½ years	Longer phrases and short sentences, with some errors and unusual constructions. Can understand much more.
3 years	Vocabulary of about 1,000 words; longer sentences with fewer errors.
4 years	Close to adult speech competence.



FIGURE 8.30 Some overeager parents try to teach their children language at a very early age. The child may enjoy the attention, but the activity is unlikely to accelerate the child's progress in language development.

their parents what words to use for certain concepts. Indeed, Deacon (1997) makes the point that languages in general evolve to become easy for infants to learn.

By age 1½, most toddlers can say a few words. (The mean is about 50.) However, they seldom link words together. Thus, a toddler who can say “Daddy” and “bye-bye” may still be unable to say “Bye-bye, Daddy.” In context, parents can usually discern considerable meaning in these single-word utterances. *Mama* might mean, “That’s a picture of Mama,” “Take me to Mama,” “Mama went away and left me here,” or “Mama, I’m hungry.”

Some toddlers follow a pattern of language development that is different from the usual one (Nelson, 1981). Instead of speaking one word at a time and learning the names of objects, these children speak poorly articulated, compressed phrases, such as “Do-it-again” or “I-like-read-Goodnight-Moon.” At first, these expressions are so poorly pronounced that adults may not realize the child is doing anything but babbling. Children who start off by generating these complex requests and phrases tend to continue doing so (Nelson, Baker, Denninger, Bonvillian, & Kaplan, 1985).

By age 2, children start producing “telegraphic” phrases of two or more words, including combinations such as “more page,” “allgone sticky,” and “allgone outside,” to indicate “read some more,” “my hands are now clean,” and “someone has closed the door,” respectively. Note the originality of such phrases; it is unlikely that children learned them by imitating their parents!

By age 2½ to 3 years, most children are generating full sentences, though each child maintains a few peculiarities. For example, many young children have their own rules for forming negative sentences. A common one is to add *no* or *not* to the beginning or end of a sentence, such as, “No I want to go to bed!” One little girl formed her negatives just by saying something louder and at a higher pitch; for instance, if she shrieked, “I want to share my toys!” she really meant, “I do *not* want to share my toys.” Presumably, she had learned this “rule” by remembering that people screamed at her when they told her not to do something. My son Sam made negatives for a while by adding the word *either* to the end of a sentence: “I want to eat lima beans

either.” Apparently, he had heard people say, “I don’t want to do that either” and had decided that the word *either* at the end of the sentence made it an emphatic negative.

At this same age, children act as if they were applying grammatical rules. (I say “as if” because they cannot state the rules. By the same token, baseball players who anticipate exactly where a fly ball will come down act “as if” they understood complex laws of physics and calculus.) For example, a child may learn the word *feet* at an early age and then, after learning other plurals, abandon it in favor of *foots*. Later, the child begins to compromise by saying “feets,” “footses,” or “feetses,” before eventually returning to “feet.” Children at this stage say many things they have never heard anyone else say, such as “the mans comed” or “the womans goed and doed something.” Clearly, they are applying rules for how to form plurals and past tenses, although they *overregularize* or *overgeneralize* those rules. My son David invented the word *shis* to mean “belonging to a female.” He had apparently generalized the rule “He—his, she—shis.” Note that all these inventions imply that children are learning rules, not just repeating word combinations.

CONCEPT CHECK

- At what age do children begin to string words into novel combinations that they have never heard anyone say before? Why do psychologists believe that even very young children learn some of the rules of grammar? (Check your answers on page 311.)

Children Exposed to No Language or to Two Languages

If children weren’t exposed to any language at all, would they make up one of their own? In rare cases, an infant who was somehow separated from all humans has grown up in a forest until discovered a few years later. Those children not only failed to show a language of their own but also failed to learn much of human language after they were given the opportunity (Pinker, 1994). However, we have no way of knowing for sure whether those children tried to develop a language when they were younger, or whether they would have tried, if they had had companions capable of responding to them.

The closest we can come to duplicating such an experience is the unfortunate condition of some deaf children. A child who cannot hear well enough to learn speech and who never sees anyone using sign language is effectively isolated from all language. The results are consistent: Those children make up their own sign language. As they grow older, they make the system more complex, linking signs together into sentences with fairly consistent word order and grammatical rules—for example, “Mother, twist open the jar, blow a bubble, so I can clap it” (Goldin-Meadow, McNeill, & Singleton, 1996; Goldin-Meadow & Mylander, 1998). Most deaf children manage to teach their system, or at least part of it, to one or both parents, although the parents do not become as fluent as the children. Deaf children who

spend much time together adopt each other's signs and eventually develop a unified system.

Although each deaf child invents a different system, most systems share some interesting similarities. For example, most include some sort of marker to indicate the difference between a subject that is doing something to an object ("the mouse eats the cheese") and a subject that is doing something without an object ("the mouse is moving"). The sign languages invented by children in Taiwan resemble those of children in the United States, even though the hearing adults in those countries are using languages with very different grammars—Chinese and English (Goldin-Meadow & Mylander, 1998).

Some children grow up in a **bilingual** environment, *learning two languages about equally well*. Bilingualism is especially common among immigrant children, who are generally bicultural as well, learning both their parents' customs and those of their new country. The representation of language in the brain is almost the same in bilingual people as it is in those who speak only one language. The research results clearly indicate that bilingual language capacity is neither more nor less lateralized to the left hemisphere than is a single language (Paradis, 1990; Solin, 1989). In the frontal cortex, in and around Broca's area, bilingual people use slightly different areas when speaking in one language or the other, but the difference is even smaller among people who have been bilingual since infancy (Kim, Relkin, Lee, & Hirsch, 1997).

If the brain representations are so similar, how do bilingual people keep their two languages separate? The answer is that they do not, at least not completely (Bijeljac-Babic, Biardeau, & Grainger, 1997). For example, one study asked people to say the ink color of various English and Spanish words, saying them all in English. You can try this task yourself with the following display:

TRY IT
YOUR-
SELF

YELLOW GREEN RED BLUE VERDE AZUL AMARILLO ROJO

If you speak only English, you probably had more trouble with the first four than with the second four. However, if you knew that *verde*, *azul*, *amarillo*, and *rojo* are the Spanish words for *green*, *blue*, *yellow*, and *red*, respectively, then you probably had some difficulty with the second four as well, demonstrating the *bilingual Stroop effect*. The better your knowledge of Spanish, the more difficult those last four items will be, even though you know you are supposed to suppress your Spanish and answer in English (Altarriba & Mathis, 1997).

Bilingualism has two disadvantages: Children take longer to master two languages than to master one (naturally), and even adult bilinguals occasionally confuse or intermix words from the two languages. The primary advantage is obvious: Someone who understands two languages well can speak with more people and read a greater variety of

literature than someone who speaks only one language. The benefits are particularly obvious for someone who learns two widely used languages, such as English and Chinese.

A second advantage of bilingualism is more subtle: A bilingual person gains extra cognitive flexibility by learning that there are different ways of expressing the same idea. For example, children younger than 6 years old who speak only one language show a strong belief that every object has one and only one name. For example, if an adult gestures toward a cup and a gyroscope and says, "please bring me the gyroscope," a child who knows the word *cup* will immediately bring the gyroscope, assuming that if one object is the cup, the other one must be the gyroscope. But the child would also fetch the gyroscope if asked to bring the *vessel*, the *chalice*, or any other synonym for *cup*. A bilingual child, however, is more likely to hesitate or to ask for clarification, showing an understanding that the unfamiliar word could refer to the cup just as easily as it could refer to the unknown object (Davidson, Jergovic, Imami, & Theodos, 1997).

CONCEPT CHECK

- 10.** Suppose you know that someone understands both English and another language, and you want to determine what that language is without even asking. How could you do this, using one of the tasks described in this section? (Check your answer on page 311.)

Understanding Language

Making sense of what we see and hear is a complex process that requires knowledge about the world. For example, consider the following sentences (from Just & Carpenter, 1987):

That store sells horse shoes.

That store sells alligator shoes.

We would not interpret the second sentence as referring to "shoes for alligators to wear," because alligators do not wear shoes. But that is a fact you had to know; the sentences themselves do not tell you that horses wear horseshoes but people wear alligator shoes. Here is another example:

I'm going to buy a pet hamster at the store, if it's open.

I'm going to buy a pet hamster at the store, if it's healthy.

Nothing about the sentence structure told you that *it* in the first sentence refers to the store and in the second sentence to the hamster. You understood because you know

that stores (but not hamsters) can be open, whereas hamsters (but not stores) can be healthy. In short, language comprehension depends on assumptions that the speaker and the listener—or the writer and the reader—share.

The context can change the meaning of whole sentences as well as individual words (Just & Carpenter, 1987). (See Figure 8.31.) For example, we generally interpret the proverb “time flies like an arrow” to mean that time passes quickly. But consider the same sentence in two other (far-fetched) contexts:

Q: You taught me one method to time the motion of cars, and a different way to time the flight of arrows. But how should I time the flight of houseflies?

A: Time flies like an arrow.

This science-fiction world has some weird insects—space flies, gravity flies, and time flies. They all eat wood, but each fly likes a different kind. Space flies like to eat a desk. Most gravity flies like chairs. Time flies like an arrow.

Language comprehension is immensely complicated, as people discover when they try to program a computer to understand language (Just & Carpenter, 1987). And yet, most of the time, we understand what we hear and read without any noticeable effort. Although many questions remain to

be answered, researchers have also made some discoveries about how comprehension takes place.

Hearing a Word as a Whole

We customarily describe the word *cat* as being composed of three sounds, *kuh*, *ah*, and *tuh*. In a sense this is misleading: The first sound in *cat* is not quite the same as the consonant sound in *kuh*; the *a* and *t* sounds are changed also. Each letter changes its sound depending on the other sounds that precede it and follow it. We cannot hear the separate letters of a word; we must hear the word as a whole.

One of the clearest demonstrations of this principle was an experiment in which students listened to a tape recording of a sentence with one sound missing (Warren, 1970). The sentence was, “The state governors met with their respective legislatures convening in the capital city.” However, the sound of the first *s* in the word *legislatures*, along with part of the adjacent *i* and *l*, had been replaced by a cough or a tone. The students were asked to listen to the recording and try to identify the location of the cough or tone. None of the 20 students identified the location correctly, and half thought the cough or tone interrupted one of the other words on the tape. They all claimed to have heard the *s* plainly. In fact, even those who had been told that the *s* sound was missing still insisted that they had heard the sound. Apparently, the brain uses the context to fill in the missing sound.

Understanding Words in Context

Many words have different meanings in different contexts. *Rose* can refer to a flower, or it can be the past tense of the verb *to rise*. Consider the word *mean* in this sentence: “What did that mean old statistician mean by asking us to find the mean and mode of this distribution?”

Just as we hear the word *legislatures* as a whole, not as a string of separate letters, we interpret a sequence of words as a whole, not one at a time. For example, suppose you hear a tape-recorded sound that is carefully engineered to sound halfway between *dent* and *tent*. If you simply hear it and are asked to say what you heard, you might reply “dent,” “tent,” or “something sort of intermediate between dent and tent.” But now, suppose you hear that same sound in context:

1. When the *ent in the fender was well camouflaged, we sold the car.
2. When the *ent in the forest was well camouflaged, we began our hike.

Most people who hear sentence 1 report the word *dent*. Most who hear sentence 2 report *tent*. Now consider two more sentences:

3. When the *ent was noticed in the fender, we sold the car.
4. When the *ent was noticed in the forest, we stopped to rest.



FIGURE 8.31 Many clever ads take advantage of the fact that a given word can have several meanings and that the readers will figure out the intended meaning based on the context.

For sentences 3 and 4, the context does not help. People are as likely to report hearing *dent* in one sentence as they are in the other (Connine, Blasko, & Hall, 1991). Think for a moment what this means: In the first two sentences, the fender or forest showed up three syllables after *ent. In the second pair, the fender or forest showed up six syllables later. Evidently, when you hear an ambiguous sound, you can hold it in a temporary “undecided” state for about three syllables for the context to help you understand it. Beyond that point, it is too late for the context to help; you hear it one way or the other and stick with your decision, even if the later context contradicts it.

Although a delayed context cannot help you hear an ambiguous word correctly, it can help you understand what it means. Consider the following sentence from Karl Lashley (1951):

Rapid righting with his uninjured hand saved from loss the contents of the capsized canoe.

If you hear this sentence spoken aloud, so that spelling is not a clue, you are likely at first to hear the second word as *writing*. That is a perfectly reasonable interpretation, until you reach the final two words of the sentence. Suddenly the phrase *capsized canoe* changes the whole scenario; now we understand that *righting* meant “pushing with a paddle.” In summary, only the immediate context can influence what you hear, but even a much delayed context can influence what you think the sentence means.

Understanding Negatives

Suppose you are trying to decide whether to buy a product at the supermarket. You notice on the package, “Contains no cyanide or rat pieces.” Do you become more likely or less likely to buy it? Less likely, if you are like most people. It is as though you did not fully believe the “no.” (After all, why would the manufacturer even mention the absence of poisons and disgusting contaminants unless there were some reason to believe these might be present?) Similarly, people who read that “Bob Talbert is

not linked with the Mafia” tend to remember the hint that he might have been linked with the Mafia (Wegner, Wenzlaff, Kerker, & Beattie, 1981).

Even if you had every reason to believe the denial, you still might act as though you did not believe it. Students in one study watched as an experimenter poured sugar into two jars. The students were then told to place whichever label they wanted on each jar. One label said “sucrose, table sugar.” The other said “not sodium cyanide, not poison.” Then, the experimenter made two cups of Kool-Aid, one with sugar from one jar and one from sugar in the other jar, and asked the students to choose one cup of Kool-Aid (Figure 8.32). Almost half the students said they had no preference, but of those who did have a preference, 35 of 44 wanted the Kool-Aid made from the jar marked “sucrose,” not from the one that denied having cyanide and poison (Rozin, Markwith, & Ross, 1990). The students acted as though the label “not cyanide” meant something was wrong with the sugar in that jar.

People have particular trouble understanding double negatives, such as “she is not unfriendly.” The state of Illinois gives the following instructions to the jurors in a murder case; note the four (!) negatives in this one sentence (emphasis added):

If you do *not* unanimously find from your consideration of all the evidence that there are *no* mitigating factors sufficient to *preclude* the imposition of a death sentence, then you should sign the verdict requiring the court to impose a sentence *other than* death.

Does that sound clear to you? Do you think the author of these instructions was trying to make the point clear to a jury?

SOMETHING TO THINK ABOUT

Sometimes, a newspaper prints an accusation (“Joe Shmo accused of selling drugs to children”) and later admits that the information was wrong. The accused person sometimes complains that the original charge was made in the form of a prominent headline but the retraction is buried in

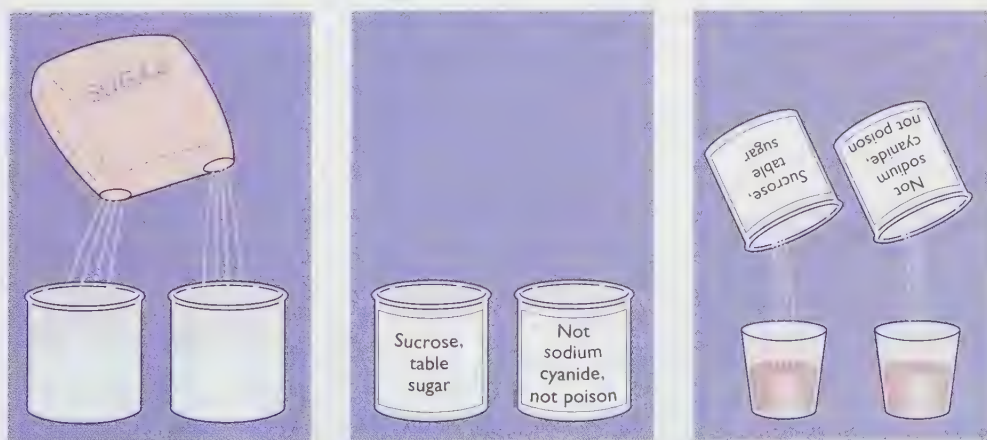


FIGURE 8.32 Most students preferred Kool-Aid made with sugar labeled “sugar” instead of sugar labeled “not cyanide,” even though they had placed the labels themselves. Evidently, people do not fully believe the word “not.” (Based on results of Rozin, Markwith, & Ross, 1990.)

the back pages where most readers overlook it. Should the accused person *want* the retraction ("Joe Shmo did not sell drugs to children") to be a prominent headline? *

Reading

As you will recall from earlier in this chapter, expertise in a chosen field, achieved after years of practice, enables one to recognize complex patterns at a glance. You may not think of yourself as an "expert" reader, but only because we usually reserve the term *expert* for someone who is far more skilled than everyone else. Nevertheless, your years of practice at reading enable you to recognize words almost instantaneously.

Consider the following experiment: The investigator flashes a single letter on a screen for less than a quarter of a second and then flashes an interfering pattern on the screen and asks, "What was the letter, C or J?" Then the experimenter flashes an entire word on the screen for the same length of time and asks, "What was the first letter of the word, C or J?" (Figure 8.33). Which question do you think would be easier to answer? Most people can *identify the letter more accurately when it is part of a whole word than when it is presented by itself* (Reicher, 1969; Wheeler, 1970). This is known as the **word-superiority effect**.

In a follow-up experiment, James Johnston and James McClelland (1974) briefly flashed words on the screen and asked students to identify one letter (whose position was marked) in each word (Figure 8.34). On some trials, they told the students to focus on the center of the area where the word would appear and to try to see the whole word. On other trials, they showed the students exactly where the critical letter would appear on the screen and told them to focus on that spot and ignore the rest of the screen. Most students more successfully identified the critical letter when they were told to look at the whole word than when they focused on just the letter itself.

The context of other letters aids recognition only if the combination is a word or something similar. For example, it is easier to recognize the difference between COIN and JOIN than the difference between C and J. But it is easier to recognize the difference between C and J than the difference between XQCF and XQJF (Rumelhart & McClelland, 1982).

FIGURE 8.33 Do you C a J? (a, b) A student watches either a word or a single letter flashed on a screen. (c, d) An interfering pattern is then flashed on the screen and the student is asked, "Which was presented: C or J?" More students were able to identify the letter correctly when it was part of a word.

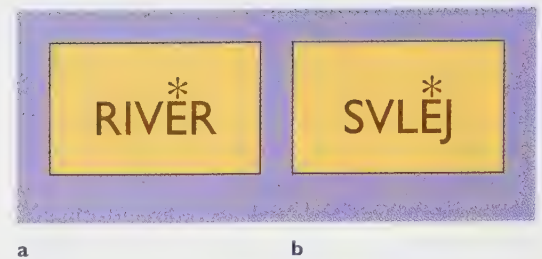
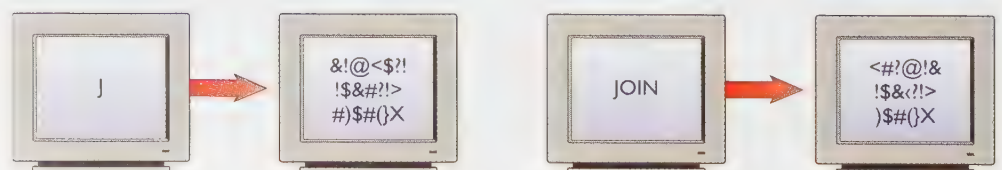


FIGURE 8.34 Students performed better at identifying an indicated letter when they focused on an entire word (a) than when they were asked to remember a single letter in a designated spot among random letters (b).

You may have experienced the word-superiority effect yourself. A common game on long car trips is to try to find every letter of the alphabet on the billboards along the way. Many people find it easier to spot a particular letter by reading whole words than by checking letter by letter.

What accounts for the word-superiority effect? According to one model (McClelland, 1988; Rumelhart, McClelland, & the PDP Research Group, 1986), our perceptions and memories are represented by vast numbers of connections among "units," presumably corresponding to neurons or sets of neurons. Each unit is connected to other units (Figure 8.35).

Each unit, when activated, excites some of its neighbors and inhibits others to varying degrees. Suppose that, at a given moment, units corresponding to the letters C, O, I, and N are moderately active—not quite active enough for a firm identification of each letter. These units excite a higher-order unit corresponding to the word COIN. Although none of the four letter units sends a strong message by itself, the collective impact is strong (McClelland & Rumelhart, 1981). This higher-level perception COIN then feeds excitation back to the letter-identifying units and confirms their tentative decisions, because they make sense in context.

Figure 8.36 is an example of the kind of phenomenon this model attempts to explain. Why do you see the top word in that figure as *RED* instead of *PFB*? After all, in the three other words of that figure, you do see those letters as *P*, *F*, and *B*. But in the top word, one ambiguous figure activates some *P* units and some *R* units; the next figure activates *E* and *F* units, and the third figure activates *D* and *B* units. All of those units in turn activate other, more complex

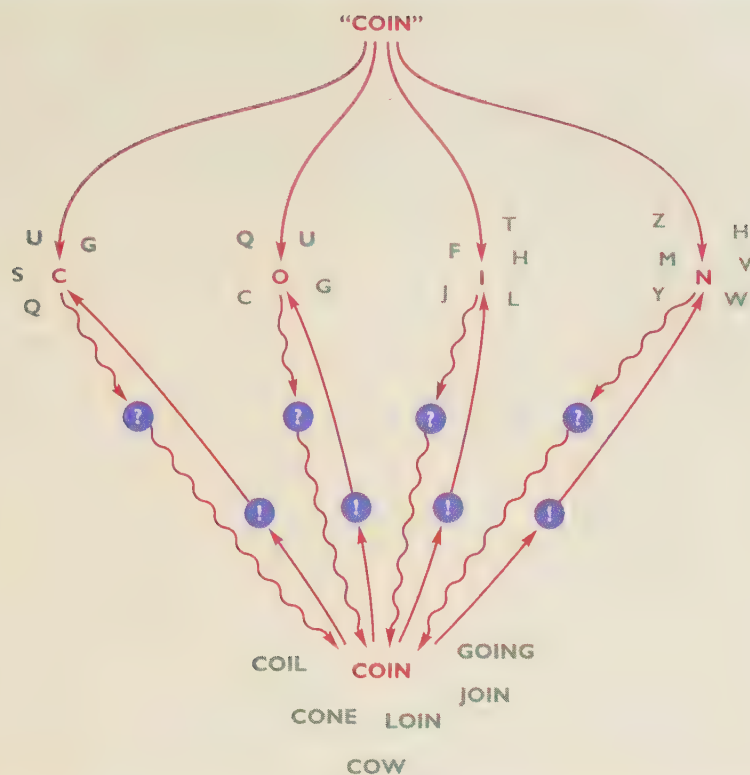


FIGURE 8.35 According to one version of the connectionist model, a visual stimulus activates certain letter units, some more strongly than others. Those letter units then activate a word unit, which in turn strengthens the letter units that compose it. For this reason, we recognize a whole word more easily than we recognize a single letter.

units corresponding to *RFB*, *PFB*, *PFD*, and *RED*. Because *RED* is the only English word in the group, the units that correspond to *RED* are easier to activate than those for *PFB* and the others. Consequently, you perceive the word as *RED*. As the *RED* unit becomes active, it in turn provides feedback to strengthen the activity of the *R*, *E*, and *D* units.

Reading and Eye Movements

In an alphabetic language such as English, the printed page consists of letters, which often form familiar clusters, which in turn form words and, finally, sentences. One kind of cluster is a **phoneme**, a unit of sound. A phoneme can be a single letter (such as *f*) or a combination of letters (such as *sh*). Another kind of cluster is a **morpheme**, a unit of meaning. For example, the word *thrills* has two morphemes (*thrill* and *s*). The final *s* is a unit of meaning because it indicates that the word is plural. (See Figure 8.37.)

CONCEPT CHECK

- 11.** How many phonemes are in the word *thoughtfully*?
How many morphemes? (Check your answers on page 311.)

When we read, do we ordinarily read one letter, one phoneme, one morpheme, or one word at a time, or do we read several words at once? And do we move our eyes steadily or in a jerky fashion? The movements are so fast that we cannot answer these questions by self-inspection.

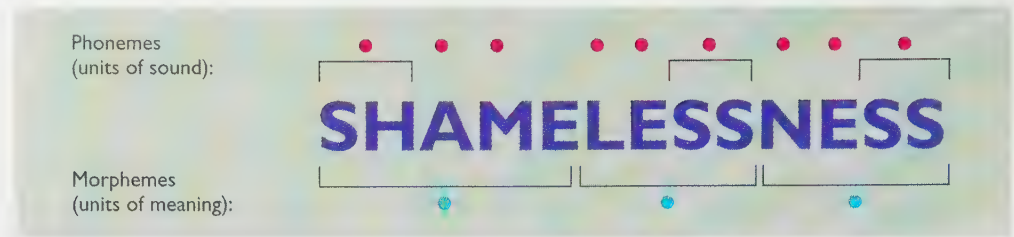
Psychologists have arranged devices to monitor people's eye movements during reading. Their first discovery was that a reader's eyes move in a jerky fashion, not steadily. You can move your eyes steadily when they are following a moving object, but when you are scanning a stationary object such as a page of print, you alternate between *periods called fixations, when your eyes are stationary*, and *quick eye movements called saccades that take your eyes from one fixation point to another*. You read during your fixations; you are virtually blind during the saccades. To illustrate that point, try the following demonstration: Look at yourself in the mirror and focus on the image of your left eye. Then move your focus to the right

**TRY IT
YOUR-
SELF**



FIGURE 8.36 A pattern across letters enables us to identify a word; word recognition in turn helps to confirm the letter identifications. Although each of the letters in the top word is ambiguous, a whole word—*RED*—is perceived. (From Rumelhart, McClelland, & the PDP Research Group, 1986.)

FIGURE 8.37 The word *shamelessness* has nine phonemes (units of sound) and three morphemes (units of meaning).



eye. Can you see your eyes moving in the mirror? (Go ahead; try it.) People generally agree that they cannot see their eyes move; that is, they are virtually blind during the eye movement.

"Oh, but wait," you say. "That slight movement of my eyes in the mirror was simply too quick and too small a movement to be seen." Wrong. Try again, but this time get someone else to look at your left eye and then shift his or her gaze to your right eye. Now you do see the other person's eye movement, so the movement itself is not too fast or too small to be seen. Go back and try your own eyes in the mirror again and observe the difference. You can see someone else's eyes moving, but you cannot see your own eyes moving in the mirror.

There are two reasons for this: First, certain areas in the parietal cortex monitor impending eye movements and send a message to the primary visual cortex, in effect telling the visual cortex, "The eyes are about to move, so shut down activity for a moment." Even if you are in total darkness, the visual cortex decreases its activity during saccadic eye movements (Burr, Morrone, & Ross, 1994; Paus, Marrett, Worsley, & Evans, 1995). Second, what you see at the end of a saccade interferes with what you saw during the saccade (Matin, Clymer, & Matin, 1972). That is, when you see a brief blur followed by a distinct stimulus, your brain concentrates on the distinct stimulus and ignores the blur.

An average adult reading a newspaper or magazine article fixates on each point for an average of about 200 milliseconds (ms). Good readers generally have shorter fixations than poor readers do (Just & Carpenter, 1987), and everyone has shorter fixations on familiar words like *girl* than on difficult words like *ghoul*. After each fixation, the saccade lasts about 25–50 ms. Thus, a typical reader has about four fixations per second.

(In the attentional blink experiments discussed earlier, a letter that is flashed on the screen interferes with another letter that is flashed 200 ms later. So why doesn't one fixation interfere with another one when you are reading? The main explanation is that the flashes are very brief in the attentional blink experiments. The other explanation relates to the word-superiority effect: It is easier to read whole words, especially in the context of a meaningful sentence, than to identify isolated letters.)

How much can a person read during one fixation? Many people have the impression that they see quite a bit of the page at each instant. Like many informal impressions, this one is wrong. Research indicates that we generally read only about 11 characters—one or two words—at a time. Recall

from Chapter 4 that human vision has its greatest acuity in the fovea, an area in the center of the retina. You can read letters when you fixate them in the fovea, but your ability to read drops sharply in the surrounding area. To demonstrate this phenomenon, focus on the point marked by an arrow (↓) in the sentences below.

TRY IT YOURSELF

1. This is a sentence with no misspelled words.
2. Xboc tx zjg rxunce with no mijvgab zucn.
3. Xboc tx zjg rxuhnj with zw cjvvgab zucn.

If you permit your eyes to wander back and forth, you quickly notice that sentences 2 and 3 are mostly gibberish. But as long as you dutifully keep your eyes on the fixation point in sentence 2, it may look all right. You can read the letter on which you fixated plus about three or four characters (including spaces) to the left and about seven to the right. This is enough for you to see—*ce with no m*—, or possibly—*nce with no mi*—. You cannot see the more distant letters well enough to be sure whether or not they spell out real words. In sentence 3, however, you do notice something wrong. Even while keeping your eyes carefully fixated on the *i* in *with*, you can see that the next word is the nonsense combination *zw* and that the previous word ends with *j*, unlike any English word you can think of.

In a study documenting this phenomenon, college students read text on a computer screen while a machine monitored their eye positions and fed that information to the computer. The computer correctly displayed whatever word a student was fixating on, plus a certain number of letters on each side of it. Beyond those letters, the computer displayed only gibberish. Students very rarely noticed that anything was unusual, unless the computer started displaying gibberish letters closer than three characters to the left or seven characters to the right of the fixation point (Underwood & McConkie, 1985). Apparently, we read in a "window" of about 11 letters—the fixated letter, about three characters to its left, and about seven to its right.

In many cases, that window includes one word plus a fragment of the next word. For example, suppose you have fixated on the point shown by an arrow in the sentence below:

↓
The government made serious mistakes.

Because of the research just described, we know that readers can see the word *serious* plus about the first three letters

of *mistakes*. Three letters do not identify the word *mistakes*; from what the reader knows, that next word could be *misspellings*, *misbehavior*, *missiles*, *mishmash*, or any of a number of other *mis*-things that a government might make. Does the “preview” of the next word facilitate reading? Yes. In one study, college students again read passages on a computer screen while a machine monitored their eye movements. The computer correctly displayed the word the student fixated on plus the next zero, three, or four letters. So the display might look like this:

↓
The government made xxxxxxx xxxxxxxx.
↓
The government made serious xxxxxxxx.
↓
The government made serious mistakes.

or like this:

↓
The government made serxxxx xxxxxxxx.
↓
The government made serious misxxxxx.
↓
The government made serious mistakes.

Students who could preview the first three or four letters of the next word read significantly faster than those who could not (Inhoff, 1989). Evidently, we do not read just one word at a time. While we are reading one word, we are previewing the next.

Would it be possible to learn to read more letters in each fixation? Perhaps, but doing so might not be helpful. Some people do read more letters per fixation, especially to the right of the fixation point. Far from gaining an advantage, such people suffer from *dyslexia*, a specific impairment of reading (Geiger, Lettvin, & Zegarra-Moran, 1992). They see too many words at once, thus confuse them, and consequently have trouble identifying any of them.

Some people read much faster than others; for example, you might wonder what “speed readers” do differently from normal readers. An average adult reader has about four or five fixations per second with occasional backtracks, for an overall rate of about 200 words per minute. Speed readers have more and briefer fixations and fewer backtracks and can double or triple their reading speed while maintaining normal comprehension. However, those who claim that they can increase their reading speed to five or ten thousand words per minute must be skipping over many of the words altogether. Saccadic eye movements last 25 to 50 ms by themselves, and the results clearly indicate that reading does not occur during saccades. Thus, it would be impossible to exceed 20–40 fixations per second, even if each fixation lasted no time at all! Given that each fixation identifies one or two words, depending on the length of the word, the theoretical maximum would be 20–80 words per second, or 1,200–4,800 words per minute. And, remember, this calculation unrealistically assumes a fixation time of



Reading is a complex skill that includes many stages, from eye movements through understanding and using the material. Investigators find that they cannot separate these stages, though how well a reader understands the material influences the speed of the eye movements. By studying reading, psychologists hope to improve methods of teaching reading.

zero. People who read faster than about 600 words per minute are actually reading some of the content and skipping over and guessing about the remainder. That strategy works well enough for fairly predictable content (like a James Bond novel), but don't try it with your chemistry textbook (Just & Carpenter, 1987).

CONCEPT CHECKS

12. Why can we sometimes read two or three short words at a time, whereas we need a saccade or two to read the same number of longer words?
13. If a word is longer than 11 letters, will a reader need more than one fixation to read it? (Check your answers on page 311.)

Reading and Understanding in Context

Good reading requires understanding word meanings. Some English words, such as *bachelor* and *uranium*, have only one meaning, so understanding them poses no special problems. Other English words, however, have multiple meanings, and we have to discern the correct one in context. For example, consider this sentence: *A pelican dove into the water and scared away the dove.* The word *dove* is potentially ambiguous, but the context enables you to infer, so fast that you were unaware of what you were doing, that the first *dove* rhymes with *stove* and means “plunged”; the second *dove* rhymes with *love* and means a “pigeonlike bird.” Now, consider another sentence: *A flying fish jumped out of the water and scared the pigeon and dove away.* In this case, the word *dove* is hopelessly ambiguous. The sentence might mean that the flying fish scared two birds (a pigeon and a dove), or that the fish scared the pigeon and then dove (plunged) away.

Good writers try to avoid severe ambiguities of this type; however, the context of a sentence is usually sufficient to clarify the meaning of each word. In some cases, a word appears early in a sentence and the context that explains it occurs later. Consider the italicized words in each of the following sentences:

**TRY IT
YOUR-
SELF**

The *table* that was full of errors was an embarrassment to the owners of the furniture store.

The *table* that was full of errors was an embarrassment to the publishers of the mathematics text.

When they chose the *lead*, they were very careful, because they wanted the new play to be a success.

When they chose the *lead*, they were very careful, because they wanted to produce high-quality pencils.

In cases such as these, most readers have a fairly long fixation time when they read an ambiguous word such as *table* or *lead*. If the previous context has made it clear which meaning is correct, a good reader almost immediately selects the correct meaning and discards the other possibilities. If the previous context provided no help, good readers keep in mind several possible meanings of the ambiguous word until later in the sentence. They show long fixation times on later words (such as *furniture* or *mathematics*, *play* or *pencils*) that clarify the meaning of the ambiguous word (MacDonald, Pearlmutter, & Seidenberg, 1994; Sereno, Pacht, & Rayner, 1992).

Poor readers are less efficient at selecting the correct meaning. Even after the context has established the correct meaning of a word, poor readers continue to keep in mind both of the possible meanings (Gernsbacher, 1993; Gernsbacher & Faust, 1991). For example, if people read the sentence “he dug with the spade,” and then an experimenter asks whether the sentence included anything related to the word “ace,” good readers promptly say no, but poor readers hesitate and sometimes say yes. Evidently, good readers suppressed the irrelevant meaning of *spade* as a playing card, but the poor readers suppressed it more slowly or less completely.

THE MESSAGE
*Language as a Defining
Feature of Humans*

At the start of this module, we considered the question, “If language is so extremely useful to humans, why haven’t other species evolved at least a little of it?” None of the research directly answers this question, but we can speculate. Certain adaptations are much more useful on a large scale than on a small scale. For example, skunks survive because they are very stinky; it wouldn’t do another species much good to be just a little bit stinky. Porcupines survive because they have long, sharp quills; having a few short quills might be slightly helpful, but not very. Similarly, a little language

might be slightly helpful (like a little stink or a few short quills), but better developed language is so enormously useful that a little bit of language development is probably an unstable condition, evolutionarily speaking. Once a species, such as humans, had evolved a little language, those individuals with still better language abilities would be at a selective advantage compared to the others. (However, of course, we are still faced with the ultimate question: Why has no other species ever evolved that first little bit of language?)

S U M M A R Y

★ **Language productivity.** Human languages enable us to create new words and phrases to express new ideas. (page 297)

★ **Language training in nonhumans.** Bonobos, and to a lesser extent other chimpanzees, have shown an ability to learn certain aspects of language. Human evolution evidently elaborated on potentials found in our ape-like ancestors but developed that potential well beyond the level it reached in other species. (page 298)

★ **Rapid language learning in children.** Children learn language at such an amazing rate, considering the unsystematic training they receive, that many psychologists believe humans are born with a predisposition to learn language. However, exactly *what* we are born with is controversial; it is hard to believe that we are born with a predisposition to learn the grammar of a particular language. (page 299)

★ **Brain organization and aphasia.** Brain damage, especially in the left hemisphere, can impair people’s ability to understand or use language. However, the brain areas that contribute to language are diffuse, and the exact brain organization varies from one person to another. (page 300)

★ **Stages of language development.** Children advance through several stages of language development, probably reflecting maturation of brain structures necessary for language and not just the total amount of experience. From the start, children’s language is creative and shows an effort to understand and use rules of grammar. (page 301)

★ **Children exposed to no language or to two.** If deaf children of hearing parents are not exposed to language, they invent a sign language of their own. Children in a bilingual environment sometimes have trouble keeping the two languages separate but gain the ability to converse with many people and, in certain ways, show greater than usual cognitive flexibility. (page 302)

★ **Understanding language.** Much of speech is ambiguous; we understand words and sentences in context by applying the knowledge we have about the world in general. (page 303)

★ **Reading.** When we read, we have fixation periods separated by eye movements called saccades. We read during the fixations, not the saccades. Even good readers can read only about 11 letters per fixation; people increase their speed of reading by increasing the number of fixations per second. (page 306)